

Word Order Variation in Spoken and Written Corpora

A Cross-Linguistic Study of SVO and Alternative Orders

Background:

- Most word order generalizations rely on written corpora, overlooking speech.

Objective:

- Compare **SVO orders** in **spoken vs. written UD treebanks** across five languages (English, French, Spanish, Slovenian, Nynorsk).

Key Observations:

- Word order differs by modality.
- Greater flexibility in speech, SVO less dominant.
- Variation differs across languages.

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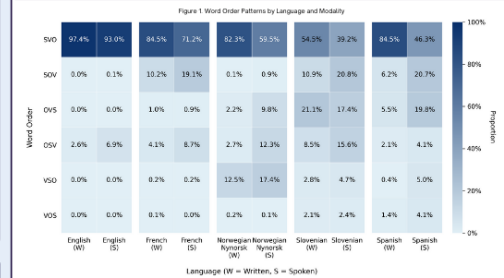
Introduction

- Word order is central in linguistic typology (WALS, Grambank) and grammar research.
- Most typological generalisations come from written data, missing modality effects (e.g., writing vs. speech).
- Recent corpus studies show word order is **gradient** and **context-sensitive**.
- Gap: Spoken word order patterns are rarely studied cross-linguistically.

Objective

- Compare **spoken vs. written** SVO word order (Subject-Verb-Object) across **five Indo-European languages** (English, French, Norwegian Nynorsk, Slovenian, Spanish).
- Research questions:
 - RQ1:** Does word order differ between spoken and written language?
 - RQ2:** If so, how does it differ?
 - RQ3:** Are these differences similar across languages?

Results and Analysis



General observations:

- Spoken vs. written difference:** Word order patterns differ between speech and writing across all examined languages.
- Greater variation in speech:** Spoken corpora show more diverse word orders, with SVO less dominant than in writing.
- Cross-linguistic flexibility:** The degree of variation differs by language (greatest in Spanish and Slovenian, smallest in English).



Figure 2: SVO Proportion Drop: Written to Spoken

Potential factors influencing the variation in speech:

- Morphological richness:** Inflection reduces reliance on fixed word order.
- Prosodic structuring:** Intonation, stress and rhythm support non-canonical sequencing.
- Interaction-specific constructions:** Frequent use of pronouns, clitics, and questions give rise to alternative patterns.
- Information structure:** Topic-focus ordering is more central in real-time production (e.g., object-fronting, postverbal subjects).

Methodology

- Data:**
 - Spoken/written UD treebanks (v2.15), matched per language.
 - Corpora: GUM (split into spoken/written subsets), Rhapsodie/GSD, NynorskLIA/Nynorsk, SST/SSJ, COSEF/GSD.
- Data extraction:** Used **STARK** tool to retrieve all clauses with finite verb + nominal subject + direct object, regardless of clause type.
- Classification:** Assigned each clause to one of six canonical word orders: SVO, SOV, VSO, VOS, OSV, OVS.
- Analysis:** Compared order distributions across spoken and written modalities for each language.

Order	Slovenian	Gloss
SVO	Mama kupače jabolka	mother buys apples
SOV	Mama kupače jabolka	mother apples buys
VSO	Kupače mama jabolka	buys mother apples
VOS	Kupače jabolka mama	buys apples mother
OSV	Jabolka mama kupače	apples mother buys
OVS	Jabolka kupače mama	apples buys mother

Table 1: Canonical Word Order Examples in Slovenian
All sentences translate as 'Mother buys apples'.

Conclusion

- Modality strongly affects word order flexibility.
- Findings challenge written-based typologies and call for integrating spoken corpora.
- Future work: more languages, in-depth analysis, finer clause type distinctions, new metrics (e.g., entropy).